



Sugarcane yield estimation in Thailand at multiple scales using the integration of UAV and Sentinel-2 imagery

Jaturong Som-ard^{1,2} · Markus Immitzer¹ · Francesco Vuolo¹ · Clement Atzberger¹

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Abstract

Timely and accurate estimates of sugarcane yield provide valuable information for food management, bio-energy production, (inter)national trade, industry planning and government policy. Remote sensing and machine learning approaches can improve sugarcane yield estimation. Previous attempts have however often suffered from too few training samples due to the fact that field data collection is expensive and time-consuming. Our study demonstrates that unmanned aerial vehicle (UAV) data can be used to generate field-level yield data using only a limited number of field measurements. Plant height obtained from RGB UAV-images was used to train a model to derive intra-field yield maps based on 41 field sample plots spread over 20 sugarcane fields in the Udon Thani Province, Thailand. The yield maps were subsequently used as reference data to train another model to estimate yield from multi-spectral Sentinel-2 (S2) imagery. The integrated UAV yield and S2 data was found efficient with RMSE of 6.88 t/ha (per 10 m × 10 m pixel), for average yields of about 58 t/ha. The expansion of the sugarcane yield mapping across the entire region of 11,730 km² was in line with the official statistical yield data and highlighted the high spatial variability of yields, both between and within fields. The presented method is a cost-effective and high-quality yield mapping approach which provides useful information for sustainable sugarcane yield management and decision-making.

Keywords Yield estimation · Sugarcane · UAV · Sentinel-2 · Earth observation · Remote sensing · Upscaling · Data augmentation

Introduction

A regular monitoring of agricultural land and crop production is crucial to obtain critical data for food security and farmland management. For example, accurate crop yield estimation provides valuable information to guide national food management, trade and

✉ Markus Immitzer
markus.immitzer@boku.ac.at

¹ Institute of Geomatics, University of Natural Resources and Life Sciences, Vienna (BOKU), Peter-Jordan-Straße 82, 1190 Vienna, Austria

² Department of Geography, Faculty of Humanities and Social Sciences, Mahasarakham University, Maha Sarakham 44150, Thailand

development strategies (Hunt et al., 2019; Weiss et al., 2020). Crop yield information is also relevant when making informed decisions on insurance, agriculture land management and market intervention (Atzberger, 2013). Nowadays, various large-scale agricultural monitoring systems offer near-real-time data on current crop conditions (Defourny et al., 2019; dos Santos Luciano et al., 2021; Rembold et al., 2019; Weiss et al., 2020; Whitcraft et al., 2015). However, mapping efforts to produce crop yield maps that describe both the within-field variability (needed by farmers for a better crop management) and the large-scale (e.g. regional or national) synoptic view remains technologically and practically challenging. A large part of this challenge is related to the cost for acquiring reference samples in the field, against which EO data can be calibrated (Emilien et al., 2021; Revill et al., 2020; Riihimäki et al., 2019). To respond to this gap, this study proposes a multi-scale workflow for yield mapping with an example for sugarcane in Thailand.

Sugarcane in Thailand

Sugarcane is a tall perennial grass in the genus *Saccharum* that grows in tropical and sub-tropical regions (Som-ard et al., 2021; Srijantr et al., 2002; Sumesh et al., 2021; Wang et al., 2020; Xu et al., 2020). Sugarcane is an economically important crop contributing to food, bio-energy production and chemical components extraction (Moraes et al., 2015; Renouf et al., 2008; Som-ard et al., 2018).

Thailand was the world's fifth-largest exporter of sugarcane in 2020 (FAO, 2022) with excellent growing conditions contributing to significant revenue (Srijantr et al., 2002). Both the government and sugar mills have promoted sugarcane expansion and developed sugarcane varieties to increase production (Sriroth et al., 2016). In Thailand, the total harvested sugarcane area has increased from 0.06 million hectares (Mha) in 1961 to 1.83 Mha in 2020, while annual production increased by 73 million tons (Mt). During the same period, the average yields increased from 32 t/ha to 41 t/ha (FAO, 2022) and sugarcane production constituted a major economical income for farmers (Chunhawong et al., 2018).

Northeast Thailand reports the highest harvested sugarcane area and production in Thailand. Farmers in this region have adopted different crop management strategies to increase production and productivity (Office of the Cane & Sugar Board, 2020; Som-ard et al., 2018; Sumesh et al., 2021). In the past years, many critical sugarcane cultivation problems arose including inaccurate yield estimation, outdated agricultural technology, sugar productivity, lack of labour, uncontrolled crop treatment, unsatisfying price decision-making process, poor irrigation management and, above all, impact of climate change (Chunhawong et al., 2018; Silalertruksa et al., 2017; Som-ard et al., 2022; Sumesh et al., 2021). These issues are addressed by government policy and related agencies to increase the efficiency for the sugar mills (Pipitpukdee et al., 2020; Sriroth et al., 2016), but not all are yet successfully tackled and solved.

Sugarcane yield estimation

An accurate and cost-efficient sugarcane yield estimation system would offer benefits to many stakeholders and is deemed necessary for price setting, to allocate quota on production by sugar mills, as well as for marketing and for monitoring the effects of drought and weather impacts.

At field scale, recent studies by Shendryk et al. (2021), Sumesh et al. (2021), Xu et al. (2020) and Yu et al. (2020) indicated that remote sensing technology can provide key

information to guide precision agricultural practices. For example, one of the important activities of sugarcane management is that farmers can use it as a cost-effective tool to better understand their field conditions and then rapidly manage water and fertilizer inputs for specific areas of the field.

Satellite-based Earth Observation (EO) data have over the past four decades strengthened crop yield estimation capabilities by providing enhanced yield mapping and up-to-date information at different spatial scales (Shendryk et al., 2021; Som-ard et al., 2021; Weiss et al., 2020). Widely used EO data includes MODIS (Aqua and Terra), Landsat, Sentinel-1 (S1), and Sentinel-2 (S2). Many studies have been published leveraging machine learning (ML) approaches for crop yield estimation (i.e., sugarcane, rice and wheat) at fine to course spatial resolutions (dos Santos Luciano et al., 2021; Franch et al., 2021; Hunt et al., 2019; Segarra et al., 2022; Son et al., 2020). Results have shown a good potential for accurate crop yield maps.

Since the launch of the first S2 in 2015, rich information (in terms of high spatial and temporal resolutions) of S2 imagery can be used to improve accurate crop yield mapping at the within-field level. Exemplary studies that have demonstrated a high potential for predicting sugarcane yield using the capabilities of S2 imagery together with ML approaches such as a random forest (RF), support vector machines (SVM) and linear regression approaches are listed in Table 1. More results and examples of EO applications for sugarcane yield estimation can be found in the recent review by Som-ard et al. (2021).

As a common denominator, all studies have reported that the lack of ground measurements is a major limitation (e.g., through crop cutting experiment), together with problems related to cloud cover conditions and difficulties to transfer models to different geographical locations (from field to national scale).

Integration of UAV data

Very high spatial resolution data derived from unmanned aerial vehicle (UAV) instruments are known to be useful for sugarcane monitoring at local scales (Sanches et al., 2018; Som-ard et al., 2021) (Table 2).

The need for reference data

The application of EO data is currently mainly limited by the scant availability of reference data. UAV data is not able to cover cost-efficiently large areas. However, very high resolution UAV-RGB data can be used to derive yield maps at the field scale based on only a few reference samples. The idea is thus to use UAV-RGB derived yield maps to augment/enlarge the number of available ground reference observations so that more reference samples are available to construct satellite-based models (Daryaei et al., 2020; Som-ard et al., 2021; Yu et al., 2020). In this case, UAV data can be considered as a proxy of ground data to train EO-based models and therefore to achieve a larger spatial coverage (Emilien et al., 2021; Revill et al., 2020; Riihimäki et al., 2019). Till now, upscaling to provide data interoperability (i.e., UAV-RGB and S2 imagery) has not been applied for sugarcane yield estimation.

Table 1 Exemplary studies for sugarcane yield estimation using Sentinel-2 data

Method	Observation data	RMSE (t/ha)	Extend (ha)	Geographic location	References
• Simple linear regression	Time series data	11.33	18,800	Australia	Rahman and Robson (2020)
• Random forest regression (RFR)	Multi-temporal data	16.00	142,300	Australia	Shendryk et al. (2021)
• Gradient boosting (GB)					
• Extreme trees (ET)					
• Extreme gradient boosting (XGB)					
• Multiple linear regression (MLR)	Multi-temporal data	4.63	56.4	Brazil	Canata et al. (2021)
• Random forest regression (RFR)					
• Support vector machine (SVM)	Multi-temporal data	12.95	12,000	Ethiopia	Abebe et al. (2022)
• Pheno-metrics in linear regression	Time series data	20.40	10,000	Ethiopia	Dimov et al. (2022)
• Ordinary least squares regression (OLS)					
• Random forest regression (RFR)					

The reported root mean square error (RMSE) are not directly comparable as experimental settings vary across studies. The extend of the study was labelled “local” for areas of approx. 50 to 20 000 ha, and “Regional” for areas of approx. 20 000 to 165 000 ha

Table 2 Exemplary studies for sugarcane yield estimation at local scales using different sensors based on UAVs

Method	Sensor	Level	R ²	RMSE (t/ha)	Geographic location	References
• Simple linear regression	RGB	Pixel	0.79	Not specified	Brazil	Sanches et al. (2018)
• Multiple linear regression (MLR)						
• Yield-correlation masking	RGB	OBIA	0.91 to 0.95	Not specified	Thailand	Som-ard et al. (2018)
• Simple linear regression	RGB	Pixel	0.90 to 0.98	Not specified	Thailand	Tanut et al. (2021)
• Data mining						
• Crop growth model	RGB	Pixel	0.95	1.29	China	Yu et al. (2020)
• Data assimilation model						
• Multiple linear regression (MLR)	LiDAR	Pixel	0.56 to 0.97	1.27 to 16.32	China	Xu et al. (2020)
• Generalized linear model (GLM)						
• Random forest regression (RFR)						
• Simple linear regression	LiDAR	Pixel	0.71	Not specified	Australia	Sofonia et al. (2019)
• Multiple linear regression (MLR)						
• Ordinary least squares regression (OLS)	RGB	OBIA	0.80	4.23	Thailand	Sumesh et al. (2021)

The coefficient of determination (R²) and root mean square error (RMSE) values are not directly comparable across studies as the experimental set-ups were different (OBIA: Object-based image analysis)

Objectives

This study attempts to estimate sugarcane yield for a complex landscape in Udon Thani Province, Thailand using a combination of UAV-RGB and S2 multispectral data. Field data were collected to calibrate UAV-based models so that detailed yield maps for 20 fields can be obtained. Subsequently, the augmented/enlarged yield dataset was used for a S2-based sugarcane yield modeling at the regional scale.

The specific objectives of the study are:

- to explore the potential of a plant height information derived from UAV-RGB data to estimate sugarcane yield at smallholder fields using only a limited number of ground truth data;
- to evaluate the optimal integration of UAV-RGB and S2 data to estimate sugarcane yield and within-field variability at the landscape scale.

Material and methods

Figure 1 illustrates the workflow used to achieve the specific objectives of the study. The input data consisted of:

- UAV data to produce within-field sugarcane masks and maps of plant heights for 20 fields;
- 41 yield measurements in 20 different fields collected through field work as well as farmer reported post-harvest yields for the same 20 fields;
- S2 imagery processed to construct monthly composites of various spectral features.

Using these data, three yield models were calibrated using random forest regression (RFR):

- **Yield_{UAV}**: UAV-based yield map obtained through direct use of 41 field samples;
- **Yield_{S2}**: S2-based yield map obtained through direct use of 41 field samples;
- **Yield_{UAV_S2}**: S2-based yield map using the UAV yield map for model calibration.

The validation of each approach was carried out with yield data not used in the training process: (i) out-of-bag (OOB), and (ii) five-fold cross validation.

Study area and field measurement

Twenty conventional sugarcane fields were selected in Udon Thani Province, Northeast Thailand (between 16° 49' 38" to 18° 03' 11" N and 101° 59' 55" to 103° 41' 29" E). Udon Thani Province covers an area of 11 730 km² (Fig. 2). The area comprises highlands with rolling plains of the Phu Phan range running north to south and plains along the Lam Pao River and the Mong River. Elevation ranges from 137 to 977 m above sea level. The climate is classified as tropical semi-humid dry-savanna (Köppen climate, classification: Aw) with a mean temperature of 27.2 °C and an average annual precipitation of 1423 mm.

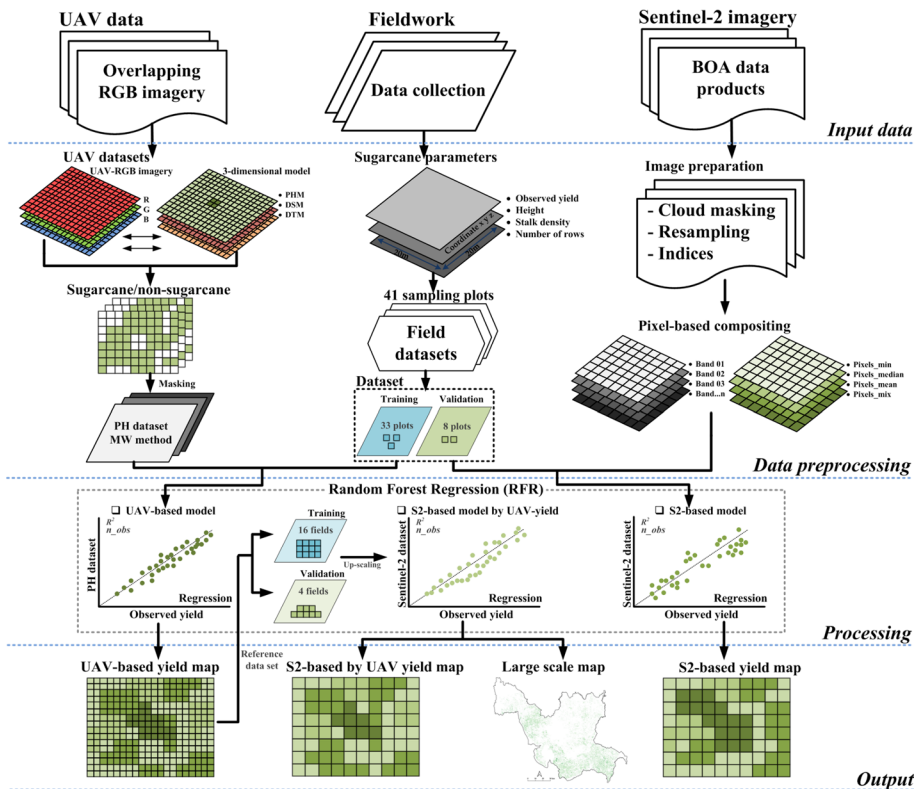


Fig. 1 Workflow diagram of sugarcane yield estimation in Thailand using the random forest regression (RFR) algorithm and integrating UAV-derived RGB and Sentinel-2 (S2) imagery

The main seasons are summer (mid-February to mid-May), rainy (mid-May to mid-October) and winter (mid-October to mid-February). This climate is highly suitable for sugarcane cultivation.

In this region, sugarcane is typically planted from September to November (rainy season) and from January to March (dry season), with a harvesting period from mid-December to March (Som-ard et al., 2022). The field survey for data collection was conducted from 15th November 2020 to 20th January 2021, with sugarcane crop cycles for only new plantations to fourth growth selected in 20 sugarcane fields. The sugarcane species, Khon-kaen-3 (KK3) is popularly planted in northeast Thailand and was used for this study. Sugarcane planting patterns are 0.2 to 0.4 m for inter-row spacing with row spacing of 1.2 to 1.5 m.

For the 20 sugarcane fields, a total of 41 sampling plots with a size of 20 m × 20 m were delineated and probed to collect sugarcane data including height, stalk density and yield. For this purpose, each sampling plot was harvested and the mentioned characteristics were measured during the fieldwork. The geographical locations of each plot were recorded using a global navigation satellite system (GNSS) receiver of UniStrong G10 from Beijing UniStrong Science & Technology company and they are displayed in Fig. 2 below, along with an exemplary field with the sampling plot locations. In addition, the farmers also provided the total yield for each field after harvesting.

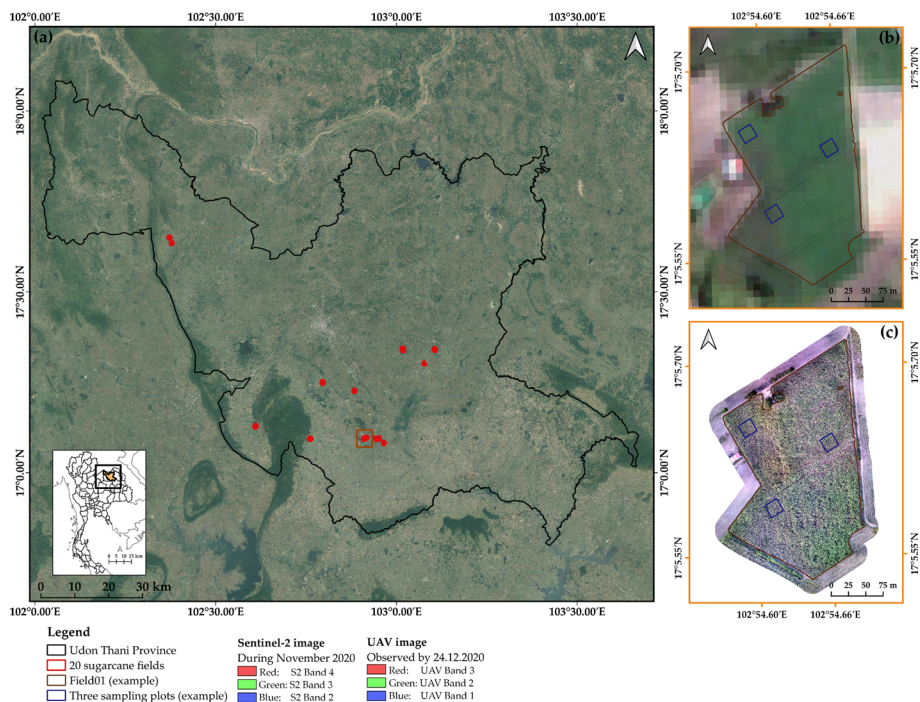


Fig. 2 The study area in Udon Thani Province, Thailand. The background (a) shows a very high spatial resolution image (source: satellite image in QGIS version 3.8). The red polygons depict the locations of the 20 sugarcane fields selected for this study. b and c show sugarcane field 01 using Sentinel-2 (S2) data, respectively, the UAV- RGB imagery. The blue polygons are the three field sample locations of sugarcane field 01. Three sampling plots (20 m×20 m) were used to collect the ground reference data in 2020

UAV data acquisition and preprocessing

The UAV data acquisition for all 20 fields was done with a Phantom 3 Professional UAV (DJI, Shenzhen, Guangdong, China) together with the field work from 15th November 2020 to 20th January 2021. Flight parameters were set as altitude 50 m, front and side overlap 80%, resulting in images with 2.1 cm spatial resolution using the DroneDeploy application (Sumesh et al., 2021). The camera operation was set for white balance as “sunny” and fixed exposure to stabilize the radiation fluctuation for RGB images (Sumesh et al., 2021). The UAV flights were performed at suitable times of the winter season (during the morning and at noon) avoiding adverse weather conditions (i.e., strong winds, rain and clouds). For each flight, 12 ground control points (GCPs) markers were used along the border of each field and geographical coordinates and elevation were recorded using a GNSS receiver together with a real-time kinematic (RTK) mode.

To create point clouds and ortho-mosaics, the UAV image data were processed in Agisoft Metashape Professional software version 1.7.5 (Agisoft LLC, 2021) using the Structure from Motion (SfM) algorithm. The ortho-mosaics and digital surface models (DSM) were exported as raster files with 2 cm pixel size, and the point clouds were exported to LAS format. The LAS file of each field was then used to generate the digital terrain models (DTM) using ArcGIS 10.8.1 software, following the processing steps of

De Souza et al. (2017). Subsequently, plant height (PH) raster data—used later as predictor variables for estimating the sugarcane yield—were obtained by subtracting the DTM from the DSM. In a totally 12 GCPs, Eight GCPs were used for georeferencing, and four GCPs as checkpoints, following previous steps from Riihimäki et al. (2019) and Sumesh et al. (2021). A detailed accuracy assessment of the UAV-SfM point cloud is shown in Table A1 (see Appendix A).

Numerous studies have shown the high potential of five green-VIs from RGB images including ground-level index (GLI) (Louhaichi et al., 2001), green–red vegetation index (GRVI) (Tucker, 1979), excess green (ExG) (Woebbecke et al., 1995), shadow index (SI) (Zhou et al., 2021) and visible atmospherically resistant index (VARI) (Gitelson et al., 2002). We used these VIs to separate sugarcane/non-sugarcane for subsequent plant height estimations of the sugarcane plants.

For each of the 20 fields, 143 to 336 reference points (sugarcane/non-sugarcane) were obtained by visual interpretation of the UAV-RGB ortho-mosaics. The reference samples were split into 80% for training and 20% for validation. Subsequently, binary masks of sugarcane vs non-sugarcane were derived for each of the 20 fields using the random forest (RF) classifier (Breiman, 2001) based on training samples of (1) UAV-RGB, (2) UAV-PH and (3) UAV-green-VI datasets as inputs. To optimize the RF-classification model, a stepwise recursive feature selection process was used based on the “Mean Decrease in Accuracy” (MDA). This improve the computations and minimize overfitting (Immitzer et al., 2018, 2019; Som-ard et al., 2022). Hyperparameters were set to their default values: (i) number of trees (*ntree*) = 1000 and (ii) number of randomly selected variables for each node (*mtry*) = square root of the number of total input features. The accuracy assessment based on the validation dataset using a confusion matrix (Congalton, 1991) and the kappa statistic (Cohen, 1960) achieved overall accuracy values from 90.7 to 98.5% (kappa 84.5 to 96.6%) for the 20 fields (see Figure A1 in Appendix A). The map results provide detailed information on the spatial distribution of sugarcane pixels, respectively, non-sugarcane pixels (i.e., soil surface, grass, trees and shrubs) (see Figure A2). The sugarcane masks were used to remove non-sugarcane areas from the PH raster data for the yield modeling as shown in Figure A2 (right).

The masked PH dataset was extracted and for all 41 filed sampling plots nine metrics (Table 3) were calculated and used as predictor variables to build a RFR model (“Yield-_{UAV}” model). The sugarcane PH dataset was extracted using “extract” function of the “*raster*” (Hijmans et al., 2015) package in R software (R version 4.1.1).

Table 3 Plant height (PH) metrics used for (UAV) high resolution yield modeling

Variable	Description
1. PH_min	Minimum height
2. PH_max	Maximum height
3. PH_range	PH.max-PH.min
4. PH_mean	Average height of each plot
5. PH_mean.squared	PH.mean*PH.mean
6. PH_median	Median height of each plot
7. PH_normalized	Normalization between zero and one
8. PH_sd	Standard deviation of height
9. PH_var	Variance of height

Table 4 The processed compositing datasets (i.e., spectral bands and indices) of Sentinel-2 imagery in November 2020 for all predictor variables as inputs, which were used for sugarcane yield estimation

Compositing method	Feature name	Predictor variable
Median	Spectral bands (10 features)	B2, B3, B4, B5, B6, B7, B8, B8A, B11 and B12
Minimum, maximum, mean and median	Spectral indices (20 features)	NDVImin, NDVImax, NDVImean, NDVImedian, GNDVImin, GNDVImax, GNDVImean, GNDVI-median, NDIImin, NDIImax, NDIImean, NDII-median, EVImin, EVImax, EVImean, EVImedian, LAImin, LAImax, LAImean and LAImedian

To consider intra-field heterogeneity, the RFR model prediction at field level was done by using a Moving Window (MW) approach. A moving window of the same size as the sampling plots was chosen. To apply the MW, the UAV data set was resampled to 1 m. For each MW the same nine PH metrics (Table 3) as for the sampling plots were calculated and the results were assigned to the center pixel resulting in layers with 1 m pixel size. The nine resulting rasters were used as input for the RFR-model to estimate sugarcane yield across the field.

Sentinel-2 dataset

In total, 38 S2 Level-2A scenes within seven S2 granules from November 2020 were used (cloud cover < 40%). The atmospheric correction was done using Sen2Cor Version 2.4 through the BOKU service (Vuolo et al., 2016). For cloud removal, the Scene Classification (SCL) band was used to exclude clouds, cirrus and cloud shadow (Zhu & Woodcock, 2012). Only the ten S2 bands with 10 m (B2, B3, B4 and B8), respectively, 20 m (B5, B6, B7, B8a, B11 and B12) spatial resolution were used for this study. All 20 m bands were resampled to 10 m.

Five different vegetation indices (VIs) were derived from the S2 images as yield predictor variables: (1) normalized difference vegetation index (NDVI) (Rouse et al., 1974), (2) normalized difference infrared index (NDII) (Klema & Smart, 1983), (3) green normalized difference vegetation index (Gitelson et al., 1996), (4) enhanced vegetation index (EVI) (Huete et al., 1997) and (5) leaf area index (LAI) (Thenkabail et al., 2000). Similar indices were used in other sugarcane yield estimation studies (Abebe et al., 2022; Canata et al., 2021; dos Santos Luciano et al., 2021; Mulianga et al., 2013; Rahman & Robson, 2020; Sanches et al., 2018).

Due to cloud cover in the study area, the median compositing (MC) method was used to construct monthly composites (i.e., November 2020) for each S2 spectral band and the selected VI datasets. This method provides high efficiency to generate spatiotemporally consistent and gap-free image composites (Griffiths et al., 2019).

In addition to the MC method, minimum, mean and maximum compositing were also applied to the VI images (Griffiths et al., 2019; Som-ard et al., 2022; Vancutsem et al., 2007). All pixel-based compositing methods were used to produce for a given VI image to select the corresponding pixels to provide VI values of each image. These compositing methods reduced the impacts of clouds and shadows when generating image datasets.

The results of each pixel-based compositing method were manually compared and checked with a very high spatial resolution image (source: satellite image in Google Earth

pro). The processed compositing datasets (i.e., spectral bands and the generated VIs) were used to construct the RFR-predictive model. Table 4 lists all predictor variables.

For the RFR model training, only the S2 10-m-pixel falling entirely into the 20 m x 20 m sampling plot was used.

Sugarcane yield modeling

For the sugarcane yield estimation we used RFR, an ensemble learning method for multiple non-linear regression problems based on uncorrelated decision trees (DTs) (Breiman, 2001). Bootstrapping with replacement randomly selects many training sets, resulting in individual DTs that are later combined using the majority vote (Breiman, 2001). RFR analysis was conducted using the *random forest* package (Liaw & Wiener, 2002) in R software.

As advantages, the RFR method provides the mean squared generalization error (i.e., RMSE and R^2) as OOB error of the RF model by managing a large number of input features and ranking their internal importance (Breiman, 2001). Recent studies have shown the high effectiveness of RFR to estimate sugarcane yield (Canata et al., 2021; dos Santos Luciano et al., 2021; Shendryk et al., 2021; Xu et al., 2020). To further improve model performance, the recursive feature elimination procedure method as “Increase of Mean Squared Error” (%IncMSE) based on the RF was used for feature selection, similar to previous studies (dos Santos Luciano et al., 2021; Immitzer et al., 2016).

The RFR model requires two hyperparameters: *ntree* and *mtry*. To find the best parameter settings we tested *ntree* values between 100 to 1,500 (15 different values) and obtained the best model performances for all models using the default value of 500. For *mtry* we used the default value of *mtry*=number of input variables divided by 3... In addition to the OOB result an additional five-fold-cross-validation was applied. The splitting was done on field level (in some fields two sampling plots were measured) and the random selection of fields and corresponding sampling plots are listed in Table 5. The same splitting was used for all model approaches.

For the sugarcane yield estimation three different model approaches were implemented:

- I. UAV-based model ($\text{Yield}_{\text{UAV}}$): The modeling was done using the yield of the 41 sampling plots and nine predictor variables (Table 3) from the PH dataset. The predictions from this model were later used to train the S2-based model ($\text{Yield}_{\text{UAV_S2}}$).
- II. S2-based model (Yield_{S2}): The yield of the 41 sampling plots and 30 S2 features (Table 4) were used to train the RFR model. This would be the “standard approach” if no UAV data would have been available.
- III. S2-based model trained by UAV yield ($\text{Yield}_{\text{UAV_S2}}$): this RFR model was based on the same 30 S2 features as in the Yield_{S2} -model, but trained on the UAV yield map ($\text{Yield}_{\text{UAV}}$) for the entire fields instead of the 41 sampling plots. To match the S2 pixel size, the UAV-derived yield map was aggregated from 1 m pixel size to 10 m (see Sect. 2.3). Note that the UAV based yields from all pixels in all fields (minimum distance to boundary: 20 m) were used to train the model.

The three RFR models were used to map sugarcane yields across 20 sugarcane fields in Udon Thani Province. The RFR-predictive models were assessed for efficiency using %IncMSE and OOB results. The optimized features for each model were ranked based on the RF method and are described in Appendix A (Table A2).

Table 5 Detailed groups of a random sampling selection for the number of sugarcane fields together with the number of sampling plots for five-fold cross-validation (Every time one group is excluded)

Samples	Five-fold dataset				
	Group A	Group B	Group C	Group D	Group E
Excluded fields (field ID)	F01, F03, F11, F17	F02, F06, F12, F16	F04, F08, F13, F14	F05, F09, F15, F19	F07, F10, F18, F20
Number of sampling plots	9	8	8	8	8

Performance of the three RFR-predictive models was assessed using the root mean square error (RMSE) and coefficient of determination (R^2). These accuracy measures were computed using (i) OOB values, and (ii) the aggregated results of the five-fold cross validation on plot level, and (iii) the aggregated results of the five-fold cross validation on field level was compared against the yield values reported by the farmers.

The best model based on S2 input data was further applied to the entire province and the generated large-scale sugarcane yield map were compared to the official yield statistics of Udon Thani Province from OCSB (Office of the Cane & Sugar Board, 2020). Average yields were calculated for each of the 20 districts of this region and compared to the official values. The provincial average yield value was multiplied by the total sugarcane areas for comparison with S2-derived production data. For estimating yield across the region, this study used sugarcane field database of the year 2020 from OCSB. This database includes a sugarcane field map which was used to produce a yield map for the entire the region based on S2 datasets and the $\text{Yield}_{\text{UAV_S2}}$ model.

Results

Estimated sugarcane yield from the UAV and Sentinel-2 datasets

Table 6 summarizes R^2 and RMSE results when using (i) solely the field reference measurements for calibration (e.g., Yield_{S2} model) vs (ii) the results when using UAV data to augment/enlarge the reference dataset (e.g., $\text{Yield}_{\text{UAV_S2}}$ model). Results clearly indicate that the upscaling of field measurements (i.e., using UAV data) to train S2 leads to a notable improvement compared to directly using the field observations and S2 data alone ($\text{Yield}_{\text{UAV_S2}}$ vs Yield_{S2}). More detailed results are reported in Table A3.

The scatterplots in Fig. 3 show the cross-validated estimated yields at plot- and field-level for the different prediction models. At plot level (a–c) ($n=41$), RMSE and R^2 values ranged between 5.25 to 9.36 t/ha for the RMSE and 0.65 to 0.90 for R^2 . As expected, the $\text{Yield}_{\text{UAV}}$ model gave the highest effectiveness for sugarcane yield estimation. Interestingly, the $\text{Yield}_{\text{UAV_S2}}$ model was clearly superior to the Yield_{S2} model, which showed the lowest accuracy amongst the three RFR models.

The validation results on field level ($n=20$) are shown in Fig. 3d–f. The $\text{Yield}_{\text{UAV}}$ model gave again an outstanding performance against the farmer reported yields, but the $\text{Yield}_{\text{UAV_S2}}$ model also provided high accuracy values. In comparison, the Yield_{S2} model trained directly on the field reference data obtained results with the lowest accuracy.

The three RFR models (i.e., $\text{Yield}_{\text{UAV}}$, Yield_{S2} and $\text{Yield}_{\text{UAV_S2}}$) were applied to map the intra-field sugarcane yield variability. Results are shown in Fig. 4 for two exemplary fields (id 01 and id 08). Estimated yields from the $\text{Yield}_{\text{UAV}}$ predictive model were smooth, with well-distributed spatial yield heterogeneity (left). The $\text{Yield}_{\text{UAV_S2}}$ model also resulted in plausible yield maps thanks to the enlarged/augmented reference data (right). On the contrary, using solely the 41 sampling plots to train the Yield_{S2} model gave unsatisfactory results (middle).

Figure 5 depicts circular bar plots displaying the average estimated yield (t/ha) from the 20 sugarcane fields along a circle from the results of the three optimized-RFR-predictive models. Figure 5a and c show similar patterns because of the use of UAV yield results to train the $\text{Yield}_{\text{UAV_S2}}$ model. Conversely, results from the Yield_{S2} -predictive model (Fig. 5b) showed markable differences compared to the other RFR models.

Table 6 Coefficient of determination (R^2) and root mean square error (RMSE) using the random forest regression (RFR) approach based on different datasets

Type of assessment	Statistics	UAV-based model trained with sampling plots ($Yield_{UAV}$)	S2-based model trained with sampling plots ($Yield_{S2}$)	S2-based model trained with UAV yield map ($Yield_{UAV_S2}$)
OOB result using all sampling data (n_{obs} for $Yield_{UAV}$ or $Yield_{S2}=41$); (n_{obs} for $Yield_{UAV_S2}=1,714$)	R^2	0.95	0.77	0.87
	RMSE (t/ha)	3.50	7.63	3.63
Plot level: five-fold cross validation (results for all sampling plots, $n_{obs}=41$)	R^2	0.90	0.65	0.84
	RMSE (t/ha)	5.25	9.36	6.88
Field level: five-fold cross validation (results for all fields, $n_{obs}=20$)	R^2	0.89	0.71	0.83
	RMSE (t/ha)	5.01	7.54	6.14

Each of the three methods (e.g., $Yield_{UAV}$, $Yield_{S2}$, $Yield_{UAV_S2}$) was assessed against three validation datasets

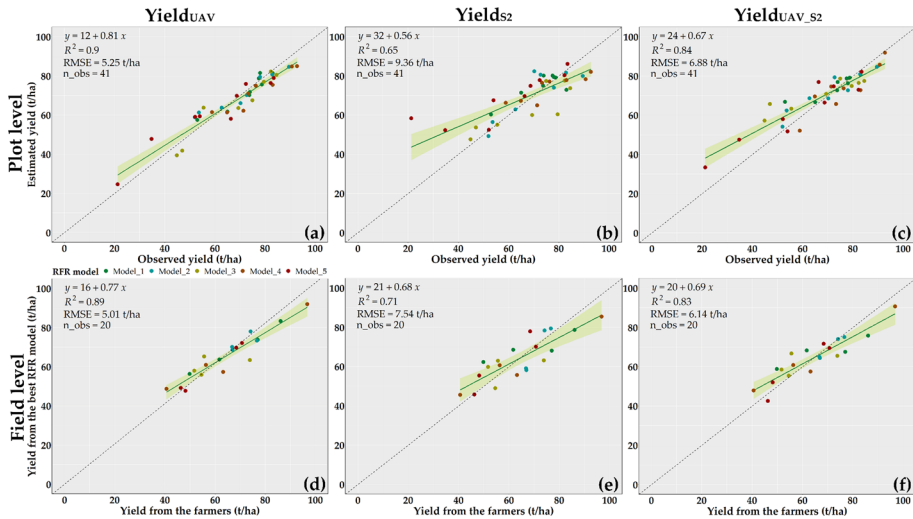


Fig. 3 Scatterplots of estimated yields obtained through five-fold cross-validation (plot level: a–c and yield by farmers (field level: d–f). Plot level: results from the UAV-based model ($Yield_{UAV}$) (a), the S2-based model ($Yield_{S2}$) (b) and the S2-based model trained by UAV yield result ($Yield_{UAV_S2}$) (c). Field level: results from $Yield_{UAV}$ (d), $Yield_{S2}$ (e) and $Yield_{UAV_S2}$ (f)

Mapping regional and within-field sugarcane yield variability across the region

The estimated sugarcane yield across the region is shown in Fig. 6. For the modeling, the $Yield_{UAV_S2}$ model was used—this model was trained using all available samples ($n=1714$). Highest yields were found in the southern areas, and lowest yield values in the northeast. Yields ranged from 41.0 to 89.9 t/ha and the mean yield value was 58.5 t/ha. The total production for the Udon Thani region was found 5.75 Mt, well in line with the provincial yield statistics from OCSB (at 5.64 Mt).

Figure 7 shows a histogram of the frequency distribution of the estimated yield at 10 m spatial resolution across the region. A positively skewed distribution can be seen, splitting the region into (large) areas with relatively low yield and (smaller) areas with high yields.

The bar chart in Fig. 8 confirms high consistency in estimating sugarcane production from 20 districts using the $Yield_{UAV_S2}$ predictive model.

Discussion

Sugarcane classification

The RF classification models achieved highly satisfactory results with excellent efficiency in sugarcane classification. The mapping results showed high overall accuracy by selecting important features. The RF classification model was also the fastest and most cost-efficient for crop mapping producers. Similarly, De Castro et al. (2018), Johansen et al. (2020) and Wang et al. (2022) indicated that UAV data together with the RF method yielded high

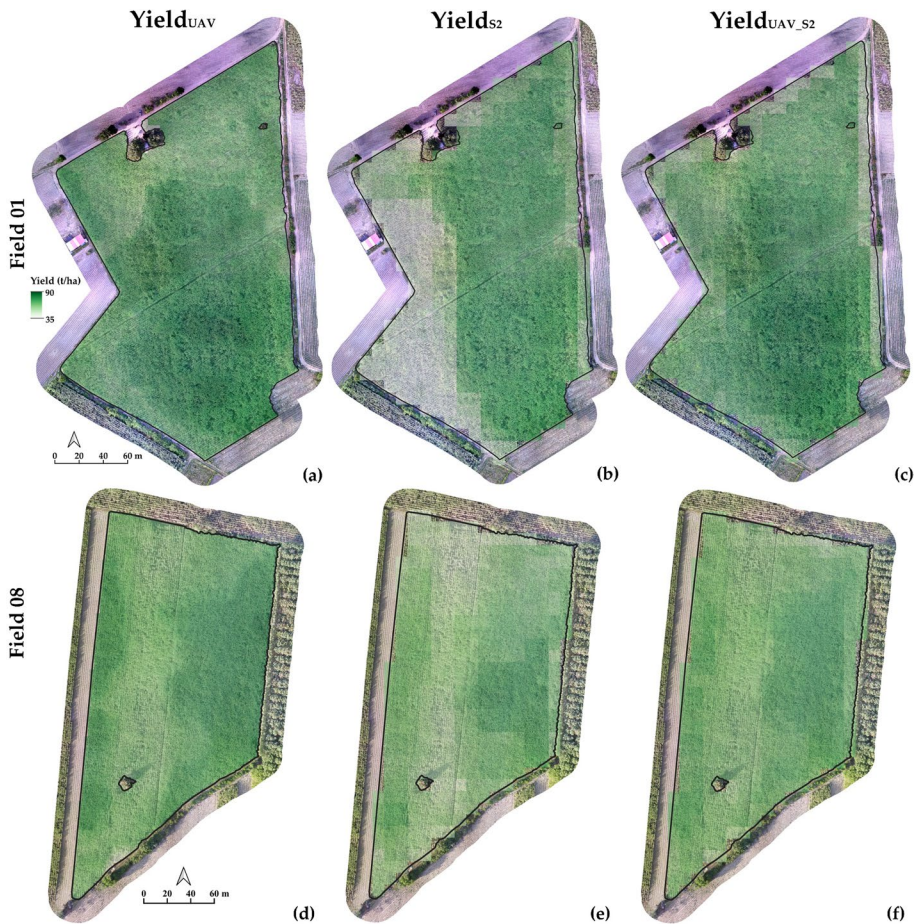


Fig. 4 Estimated yield maps (example: fields 01 and 08) from the optimized-random forest regression (RFR) models and the UAV-derived RGB and S2 image datasets. Yield maps were derived from the UAV-based model ($Yield_{UAV}$) (a and d), the S2-based model ($Yield_{S2}$) (b and e), and the S2-based model trained by UAV yield result ($Yield_{UAV_S2}$) (c and f). As background UAV- RGB images of field 01 and 08 were used

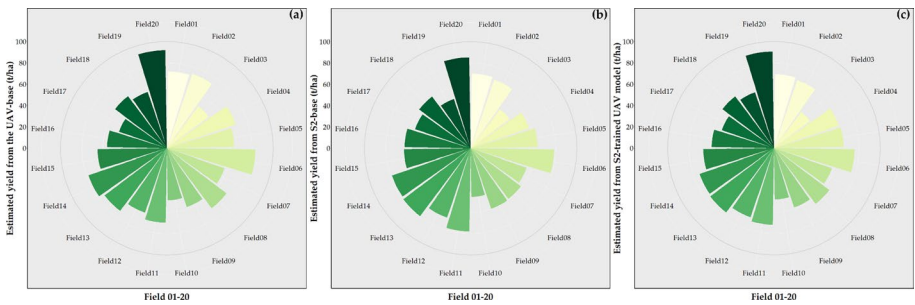


Fig. 5 Circular bar plots for the estimated yields for each of the 20 sugarcane fields. Results from the UAV-based model ($Yield_{UAV}$) (a), the S2-based model ($Yield_{S2}$) (b) and the S2-based model trained by UAV yield result ($Yield_{UAV_S2}$) (c)

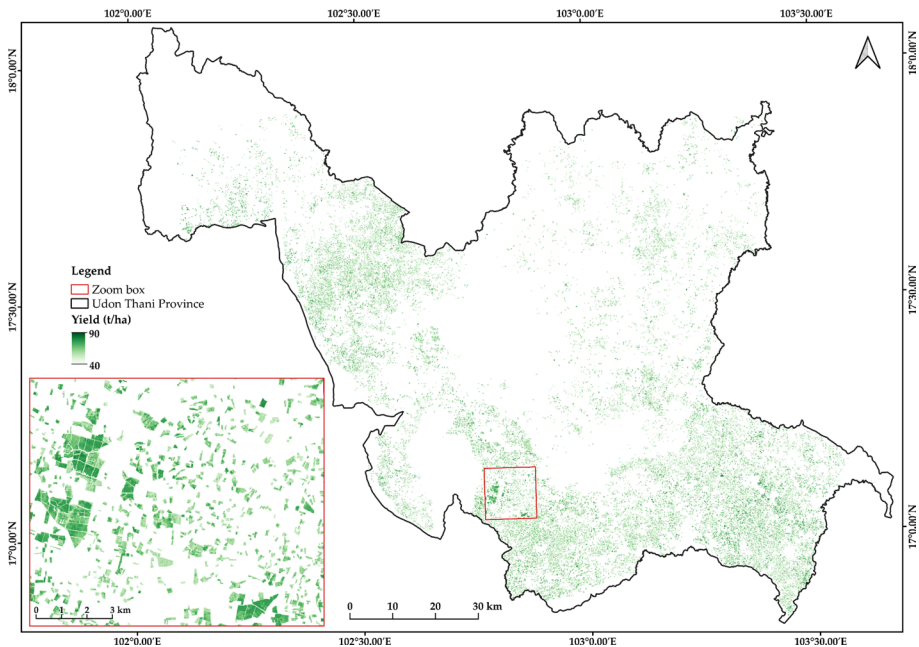


Fig. 6 Spatial distribution map of estimated sugarcane yield for the entire Udon Thani Province based on the optimized-random forest regression (RFR)-predictive model involving UAV and S2 data ($\text{Yield}_{\text{UAV_S2}}$ model)

benefits for mapping small-sized crop fields, offering efficient information as reference data to connect and train to large-scale EO-satellite data.

Estimated sugarcane yield

UAV yield results from the $\text{Yield}_{\text{UAV}}$ model were used to train the S2 datasets for constructing the $\text{Yield}_{\text{UAV_S2}}$ model. In this two-step process, UAV data is used to enlarge/augment the labor-intensive and costly field work by generating highly accurate, high resolution yield maps for the probed fields, and this expanded reference data is used to train the large-scale S2-based models. The resulting $\text{Yield}_{\text{UAV_S2}}$ -predictive model showed high potential in estimating sugarcane yield with higher accuracies (lower RMSE and higher R^2) compared to the Yield_{S2} model trained directly on the field samples, due to the augmented/expanded reference data set. This result underlines the high efficiency of UAV-derived reference data. The maps based on UAV data provide detailed information for producing vegetation or crop yield maps at multiple scales as also highlighted in recent studies (Daryaei et al., 2020; Jiang et al., 2022; Revill et al., 2020; Riihimäki et al., 2019; Zhang et al., 2019). Therefore, upscaling field measurements using UAV data showed high improvement over models directly trained on (fewer) field samples. Future studies should quantify the minimum number of field reference samples needed to generate useful UAV models.

When trained against 41 reference samples, the $\text{Yield}_{\text{UAV}}$ predictive model provided high efficiency and consistency in estimating sugarcane yield at several smallholder fields with low RMSE and high R^2 . The RFR predictive models achieved high accuracy

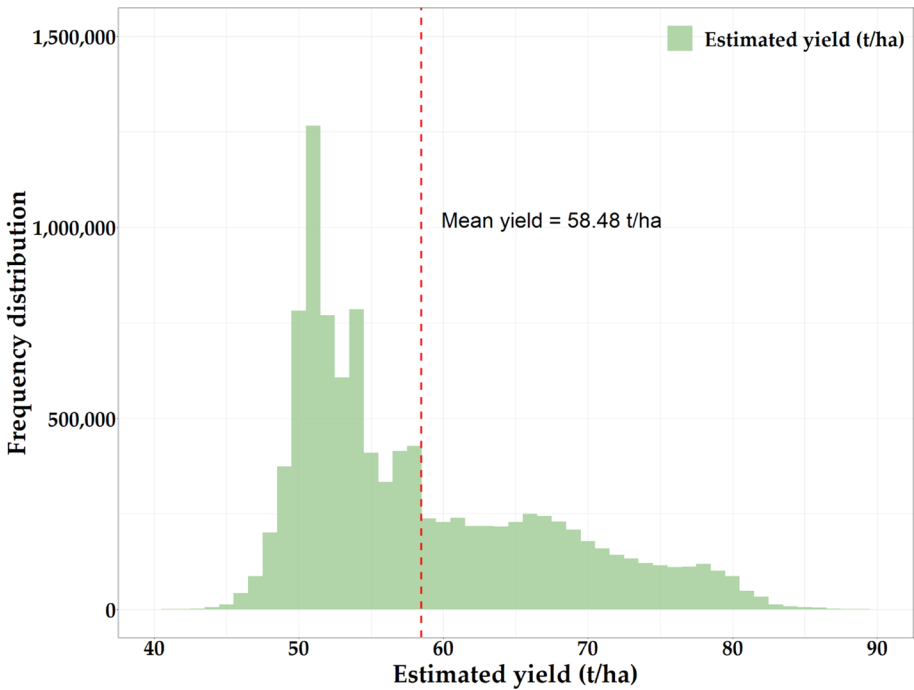


Fig. 7 Histogram showing the frequency distribution of estimated yield (t/ha) across the region from 10 m S2 datasets (bin width=1) based on the optimized-random forest regression (RFR)-predictive model trained against the UAV yield ($Yield_{UAV_S2}$ model)

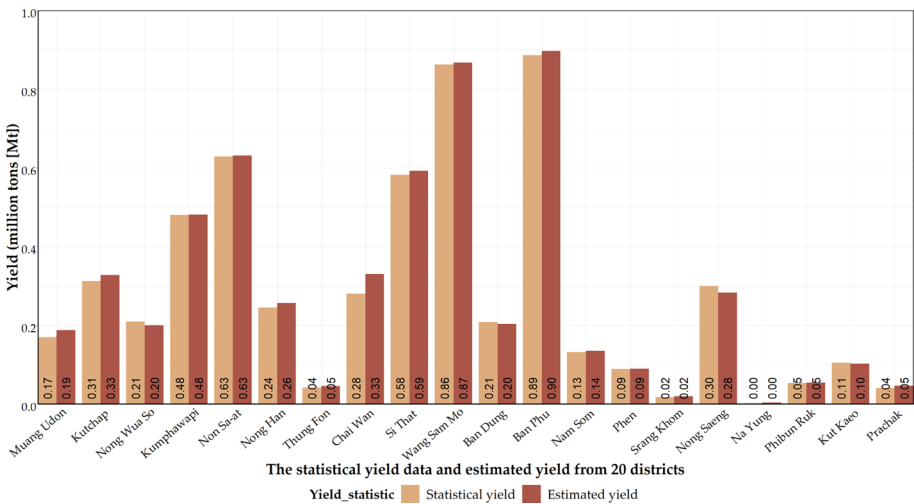


Fig. 8 Bar chart of statistical yield data (Megatonnes [Mt]) from 20 districts and estimated production based on the $Yield_{UAV_S2}$ -predictive model

($R^2 > 0.89$), compared to recent studies of (Sumesh et al., 2021; Xu et al., 2020), indicated UAV data are valuable information for sugarcane yield estimation at local scale. Results demonstrated different yield densities, strongly related to plant height. Similar to previous studies (De Souza et al., 2017; Sumesh et al., 2021; Xu et al., 2020; Yu et al., 2020), results successfully demonstrated that the PH dataset from the UAV-derived RGB improved sugarcane yield estimation compared to field measurements on a few plants. The RFR model addressed the main problems of overfitting and handling of input features for model performance (Xu et al., 2020). The estimated intra-field variability map from the $\text{Yield}_{\text{UAV}}$ model revealed high spatial yield heterogeneity (t/ha) and showed satisfactory result, which can be used to train the S2 model ($\text{Yield}_{\text{UAV-S2}}$).

The S2 dataset trained against field measurements showed good yield estimation which could however be further improved by using the larger reference dataset derived from the UAV data. The RFR method—together with S2 imagery—was in this respect appropriate for estimating sugarcane yield, similar to results found by Canata et al. (2021) and Shendryk et al. (2021). Our study had however to deal with small field sizes and a complex landscape. The obtained accuracies demonstrate a good agreement with previous studies (Canata et al., 2021; Rahman & Robson, 2020).

Our study demonstrated the great potential of integrating UAV and S2 datasets to estimate sugarcane yield. Findings confirmed that the upscaling method overcame the small size of field measurement and improved the spatial heterogeneity of yield mapping. Other studies (Daryaei et al., 2020; Revill et al., 2020; Riihimäki et al., 2019; Zhang et al., 2019) also indicated that the key advantages of UAV data could be adopted as a reference to the EO dataset for vegetation upscaling. In future studies, our upscaling yield estimation approach can be applied to predict yields for sugarcane and other crops on regional or national scales.

Regional sugarcane yield mapping

Yield models based on S2 data have the potential to produce large scale (wall-to-wall) maps. Compared to the official yield data from the provincial statistics, the S2-derived sugarcane yield map for the Udon Thani Province indicated excellent efficiency based on the $\text{Yield}_{\text{UAV-S2}}$ model (e.g., the Yield_{S2} model trained on the UAV maps). Our map show accurately the spatial distribution of inter- and intra-field yield variability. Such accurate sugarcane yield estimations could be used to complement statistical yield data with crop warning systems to inform national or international trade for Thai government policy, even for production management by local sugar mills. A spatial yield map is also a valuable information for sustainable agricultural management for sugar mills, authorities, local for increasing productivity and production. Such maps can also support farmers in better crop management practices for specific areas of the field. Moreover, such maps provide important evidence for various applications such as insurance, response and recovery after negative impacts from climate change and extreme weather events, etc. For very large regions, the additional use of meteorological variables is potentially helpful and should be further investigated [e.g., Cheng et al. (2022)]. To improve the model performance, future studies should consider other machine learning algorithms, as well as additional predictive variables such as topography, climate, or soil quality.

In this study, we obtained very good results with S2 datasets from the early harvest season (November/December), and our estimated yield map fully agreed with previous findings (Abebe et al., 2022; Mulianga et al., 2013; Shendryk et al., 2021). In future studies,

EO time series data using our proposed “augmentation method” can be used to improve sugarcane yield estimations and also support for example crop stress detection [e.g., Berger et al. (2022)].

Conclusions

The technology with respect to unmanned aerial vehicles (UAVs) and satellite sensors is constantly improving, and we demonstrate that the combined data offer highly efficient means to estimate crop and crop yield across different scales. We specifically propose to integrate UAV and Sentinel-2 (S2) imagery to estimate crop yields so that costly and time-consuming field work can be minimized.

Important findings and further suggestions are:

- in cases of insufficient field data, UAV data have a high potential to generate additional reference data to subsequently train satellite-based models (i.e. S1, S2 and Landsat data);
- our proposed “augmentation methodology” confirmed that integration of UAV and S2 datasets improved estimation of sugarcane yield across scales;
- future research should apply the proposed method to estimate yields in other regions and for other crop types;
- the RFR method showed high potential for estimating sugarcane yield. However, future studies should assess this method against other machine learning algorithms;
- our results offer valuable information to guide national food management, trade, sugar mills and government policy to promote sustainable agricultural management.

Appendix A: Supplementary data

See Table 7, 8, and 9; Figs. 9 and 10.

Table 7 Root mean square error (RMSE) assessment (in meters) from the ground control points (GCPs) and control points (CPs) for the 20 sugarcane fields

Field ID	GCP X	GCP Y	GCP Z	CP X	CP Y	CP Z
Field01	0.017	0.007	0.032	0.010	0.021	0.049
Field02	0.019	0.018	0.026	0.011	0.007	0.028
Field03	0.010	0.007	0.008	0.019	0.017	0.011
Field04	0.010	0.019	0.020	0.015	0.022	0.025
Field05	0.071	0.082	0.074	0.160	0.141	0.082
Field06	0.014	0.072	0.018	0.016	0.079	0.038
Field07	0.011	0.012	0.005	0.018	0.018	0.011
Field08	0.026	0.022	0.026	0.014	0.025	0.047
Field09	0.010	0.008	0.007	0.024	0.032	0.057
Field10	0.015	0.009	0.005	0.008	0.018	0.016
Field11	0.007	0.007	0.011	0.013	0.011	0.015
Field12	0.019	0.020	0.012	0.014	0.025	0.027
Field13	0.018	0.018	0.048	0.016	0.027	0.091
Field14	0.011	0.012	0.023	0.014	0.017	0.032
Field15	0.012	0.009	0.027	0.012	0.011	0.031
Field16	0.010	0.015	0.019	0.011	0.017	0.023
Field17	0.011	0.007	0.004	0.012	0.008	0.007
Field18	0.016	0.007	0.003	0.013	0.011	0.005
Field19	0.012	0.008	0.005	0.016	0.010	0.006
Field20	0.002	0.004	0.004	0.011	0.020	0.009

Table 8 Ranking importance of feature variables based on the three random forest regression (RFR) models with all sampling datasets and remote sensing datasets (i.e., UAV-derived RGB and Sentinel-2 imagery)

RFR model	Feature variables
1st UAV-based model (Yield _{UAV})	CHM_median, CHM_mean, CHM_mean.sq and CHM_min
2nd S2-based model (Yield _{S2})	B08A, B07, B08, LAImax, EVImedian, NDIIImedian, B06, NDIImean, NDVIImedian, EVImean, B03, LAImedian, NDIIImax, B02, GNDVIImedian, EVImin, GNDVIImax, LAImean, NDVIImax, B05, GNDVIImean, EVImax, GNDVImin, B04, B12 and B11
3rd S2-based trained by UAV-yield model (Yield _{UAV_S2})	GNDVImin, B05, B12, B03, B02, B11, EVImedian, B06, LAImin and B08

Table 9 Coefficient of determination (R^2) and root mean square error (RMSE) from OOB values and five-fold cross-validation using the random forest regression (RFR) approach based on different datasets (i.e., UAV-derived RGB and Sentinel-2 (S2) imagery) and the yield from field measurement and the UAV yield result

Type of assessment	Statistic measurement	Yield from field observation (41 sampling plots)		Yield from UAV map (all pixels)
		UAV-based model (Yield _{UAV})	S2-based model (Yield _{S2})	
1st fold—OOB (n_obs for Yield _{UAV} or Yield _{S2} = 33; (n_obs for Yield _{UAV_S2} = 1,445)	R2	0.90	0.71	0.82
	RMSE (t/ha)	4.39	9.34	5.01
1st fold—Validation (n_obs = 8)	R2	0.89	0.68	0.83
	RMSE (t/ha)	3.05	5.89	5.15
2nd fold—OOB (n_obs for Yield _{UAV} or Yield _{S2} = 33; (n_obs for Yield _{UAV_S2} = 1,177)	R2	0.95	0.77	0.84
	RMSE (t/ha)	3.56	8.02	4.03
2nd fold—Validation (n_obs = 8)	R2	0.93	0.75	0.87
	RMSE (t/ha)	4.50	6.43	6.05
3rd fold—OOB (n_obs for Yield _{UAV} or Yield _{S2} = 33; (n_obs for Yield _{UAV_S2} = 1,395)	R2	0.92	0.73	0.83
	RMSE (t/ha)	3.59	8.12	4.05
3rd fold—Validation (n_obs = 8)	R2	0.91	0.73	0.86
	RMSE (t/ha)	5.43	9.05	9.23
4th fold—OOB (n_obs for Yield _{UAV} or Yield _{S2} = 33; (n_obs for Yield _{UAV_S2} = 1,535)	R2	0.90	0.74	0.85
	RMSE (t/ha)	4.08	9.94	4.39
4th fold—Validation (n_obs = 8)	R2	0.90	0.74	0.82
	RMSE (t/ha)	5.75	7.17	5.85
5th fold—OOB (n_obs for Yield _{UAV} or Yield _{S2} = 32; (n_obs for Yield _{UAV_S2} = 1,304)	R2	0.90	0.73	0.80
	RMSE (t/ha)	3.87	8.72	5.19
5th fold—Validation (n_obs = 9)	R2	0.91	0.75	0.90
	RMSE (t/ha)	6.05	14.53	7.88

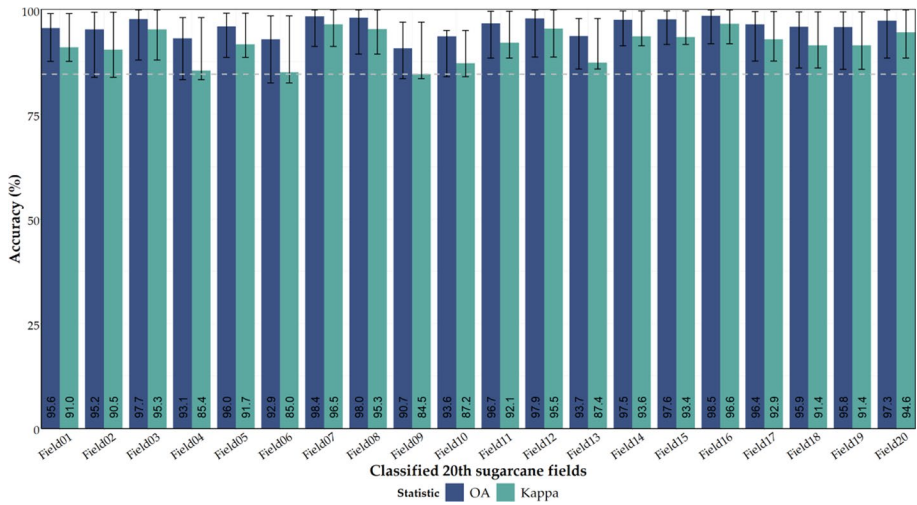


Fig. 9 Assessment of overall accuracy (OA) and kappa statistic with 95% confidence interval (CI) based on independent validation datasets for mapping sugarcane and non-sugarcane

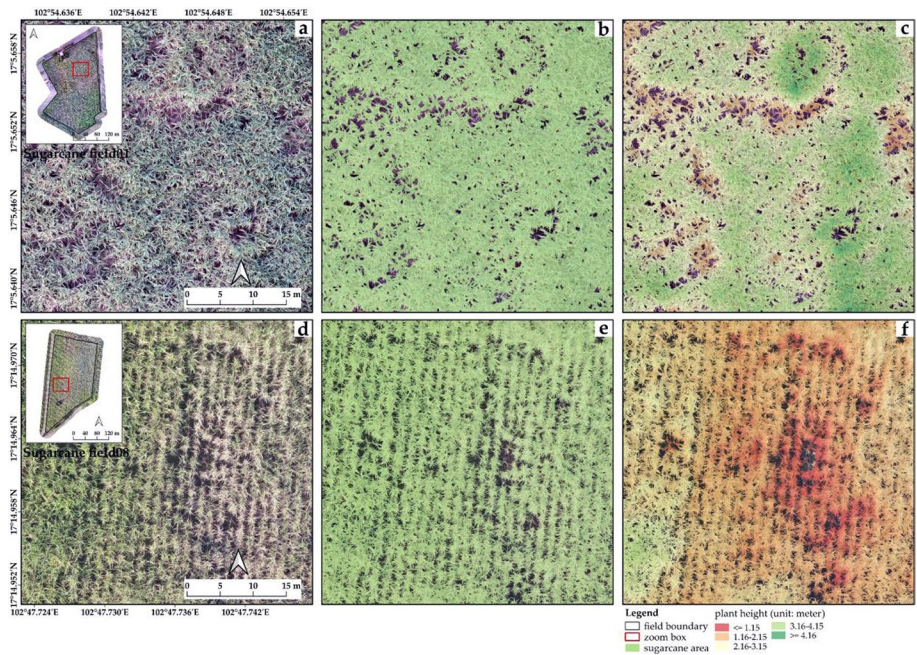


Fig. 10 Sugarcane and non-sugarcane classification for 2 sugarcane fields using the random forest (RF)-classification model based on the UAV-derived RGB dataset (i.e., UAV-RGB image, PH raster data and green-VI datasets). The example fields 01 and 08 are shown: the UAV-derived RGB image (left: **a** and **d**), sugarcane pixel (middle: **b** and **e**) and sugarcane plant height (PH) model (right: **c** and **f**)

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Author contributions JS: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, resources, software, project administration, validation, visualization, writing—original draft, writing—review and editing. MI: conceptualization, investigation, methodology, software, supervision, validation, visualization, writing—review and editing. FV: conceptualization, supervision, validation, writing—review and editing. CA: conceptualization, funding acquisition, methodology, project administration, resources, supervision, validation, visualization, writing—review and editing.

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Data availability The datasets generated during and analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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